Sequence composition of Invitrogen 1 Kb plus DNA ladder, a common microinjection stuffer DNA for *C. elegans* transgenesis.

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Abstract

In this paper, we determine the sequence of the 1 Kb plus DNA ladder from Invitrogen, a commonly used stuffer DNA for generating extrachromosomal arrays in *C. elegans* by microinjection. We sequenced the DNA ladder to improve the assembly of arrays from long-read sequencing and, more generally, to determine possible effects on transgene incorporation and expression. Sequence analysis revealed that the ladder is derived from multiple sources, including expression plasmids, lambda phage, and *E. coli* rapA. RNA-seq analysis of published data detected putative ladder-derived reads. We provide reference sequences for all 18 bands of the ladder.

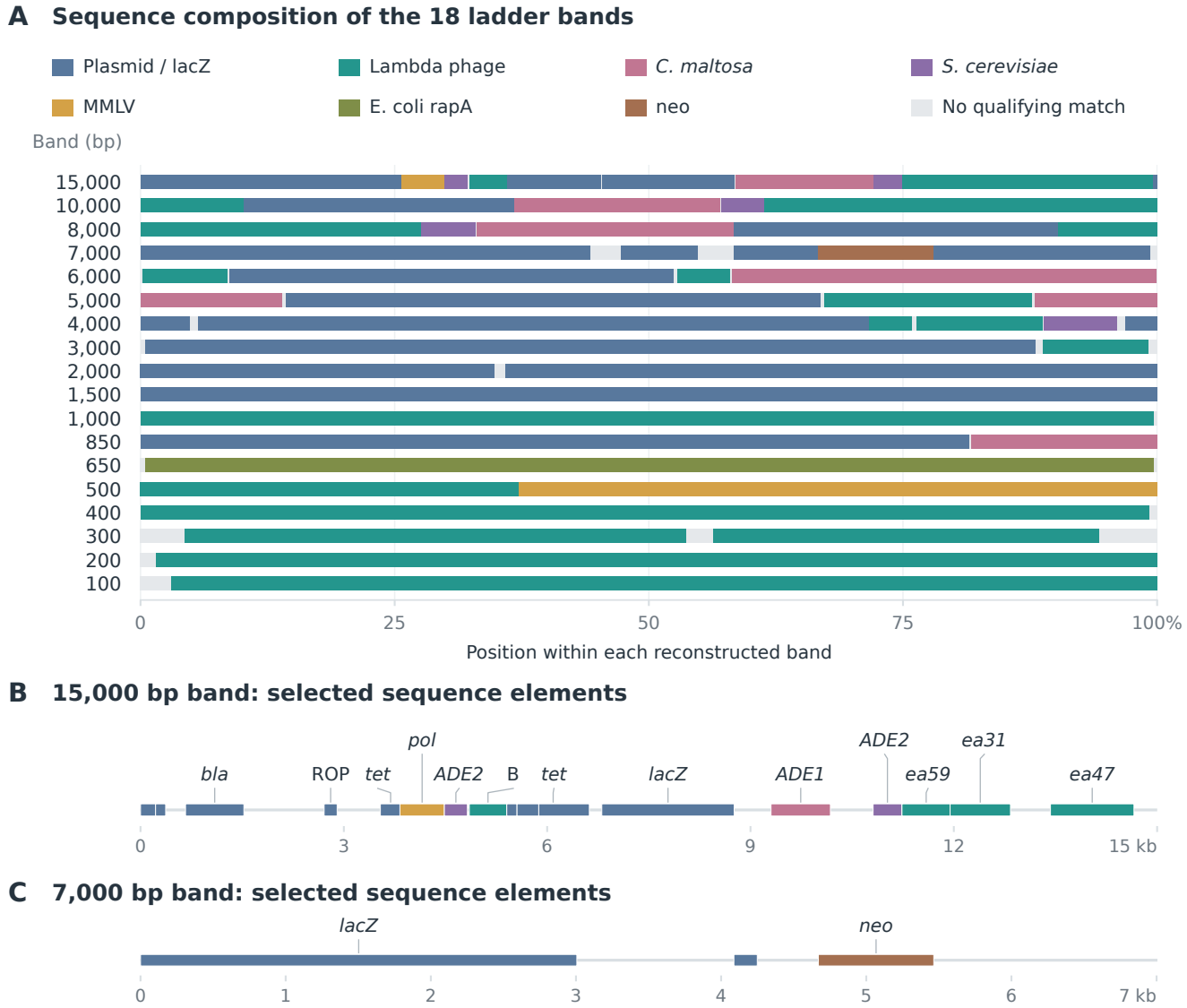


Figure 1. A. Sequence composition of each band, scaled to its own length. Colors indicate reference matches (≥95% identity; ≥50 alignment columns). Plasmid / lacZ combines pMC1871, pCMVβ, pBR322 and *E. coli lacZ* matches. Gray indicates no qualifying match. B. Annotation of the main sequence elements in the 15 kb band. C. Annotation of the main sequence elements in the 7 kb band. Annotations indicate sequence similarity. Lambda phage genes: *B*, DNA-packaging portal protein; *ea59*, AAA-family ATPase; *ea31*, DNA-cleaving HNH endonuclease; *ea47*, protein of uncharacterized function. MMLV, Moloney murine leukemia virus; ADE1, *C. maltosa*; ADE2, *S. cerevisiae*.

Description

Microinjection of DNA is a common technique used in *C. elegans* research to generate transgenic animals (Evans, 2006). The success and efficiency of this transformation can be influenced by various factors, including the concentration and composition of the injection mixture (Nance and Frøkjær-Jensen, 2019). In order to generate stable and heritable extrachromosomal arrays, homology between injected DNA fragments and a concentration of approximately 100 ng/μl is optimal (Mello et al., 1991). As transgenes are typically injected at a concentration of 5 to 10 ng/μl, "stuffer" DNA is often added to the mixture to achieve a total concentration of 100 ng/μl. A common choice for stuffer DNA is the 1 Kb plus DNA ladder from Invitrogen (cat number: 10787018) (e.g., Andersen and Horvitz, 2007; Gibson et al., 2022; El Mouridi and Frøkjær-Jensen, 2022; Prior et al., 2017; Wei et al., 2019). The sequence of this DNA ladder is not publicly available. However, the ladder is frequently a major component of extrachromosomal and integrated arrays (El Mouridi et al., 2022). Knowledge of its sequence and composition will benefit researchers using arrays, particularly for accurate assembly from long-read sequencing. In this short report, we present reference sequences for all 18 bands of this DNA ladder and describe the sequence elements it contains.

The largest band on the ladder contains sequence matching pMC1871, an *E. coli* expression plasmid for studying gene fusions (Nielsen et al., 1983), and ~4 kb of sequence matching lambda phage. The band also contains sequences matching ade1 and ade2 genes from *Candida maltosa* and *Saccharomyces cerevisiae*, respectively, and the lambda phage portal protein gene and a mouse retrovirus pol gene. All bands larger than 1 kb share sequence with the 15 kb band; the 7 kb band also shares sequence with the pCMVβ mammalian expression vector (MacGregor and Caskey, 1989). Most smaller bands match the lambda phage genome; the 650 bp band is a part of the *E. coli* rapA gene, the 500 bp band contains lambda and mouse retrovirus sequences, and the 850 bp band contains pBR322 and *C. maltosa* sequences. We analyzed published *C. elegans* RNA-seq datasets (Priyadarshini et al., 2022) for evidence of transcription from the ladder sequences. We detected reads matching ladder sequences, but these could also originate from other constructs or contaminants sharing plasmid or microbial sequences with the ladder. The read patterns did not provide convincing evidence of mature ladder-derived mRNA (data not shown).

Methods

Long read DNA sequencing: 1 μg of DNA ladder (Thermo Fisher, cat number: 10787018) was used with a ligation sequencing kit (Oxford Nanopore Technologies, cat number: SQK-LSK110) to prepare long read sequencing libraries following the manufacturer's protocol. The library was sequenced on an Oxford Nanopore MinION device. Guppy v6.5.7 was used with super-accurate settings for base calling. In order to correct read errors, reads of similar lengths were aligned and polished using minimap2 v2.24 and Racon v1.5.0, and consensus sequences corresponding to each band were generated using pyabpoa v1.5.6 (Li, 2018; Vaser et al., 2017; Gao et al., 2021). Python code to generate the consensus sequences is available at https://github.com/fffamk/DNA-ladder. Raw sequencing data and feature annotations are available at https://doi.org/10.5281/zenodo.23471697

DNA sequences of the 1 Kb Invitrogen ladder are provided in the accompanying FASTA file (ladder_sequences.fasta).

Data analysis: We queried a curated panel of 15 reference sequences with the ladder sequences using NCBI BLAST+ in order to identify their likely origin. Partial matches were used to assign sequence sources and obtain relevant annotations. The figure was generated using matplotlib software. Published RNA-seq data were screened using Bowtie2 v2.4.4.

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